#101

CS Final Project Presentation

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SmartSpend: Budget Calculator for Scholars

Project Statement

Based on our observation while roaming around the campus, some students have been complaining about having difficulty in managing their allowance properly, often spending too much on wants and unnecessary expenses. There is a need for a tool that helps them budget their money wisely and develop better spending habits.

Project Objectives

- To create a program that calculates and tracks a student's spending based on their allowance and chosen budget categories.
- Help students visualize how their money is divided among Savings, Food, Toiletries, Miscellaneous, and School needs.
- To provide a simple and user-friendly interface suitable for students with little to no experience in budgeting

Planned Features

- ·Automatic 10% Savings Deduction
- Percentage-Based Budgeting: Allows users to allocate their budget by importance rather than just numbers.
- Emergency Funds: Automatically moves unused percentages into an Emergency Fund.
- Real-time Live Update UI: Shows the user exactly how much percentage they have left while they type.
- Data Storage: Saves all calculations to a history.txt file so data isn't lost when the app closes.
- Animated Results: A cool UI experience that types out the budget summary.

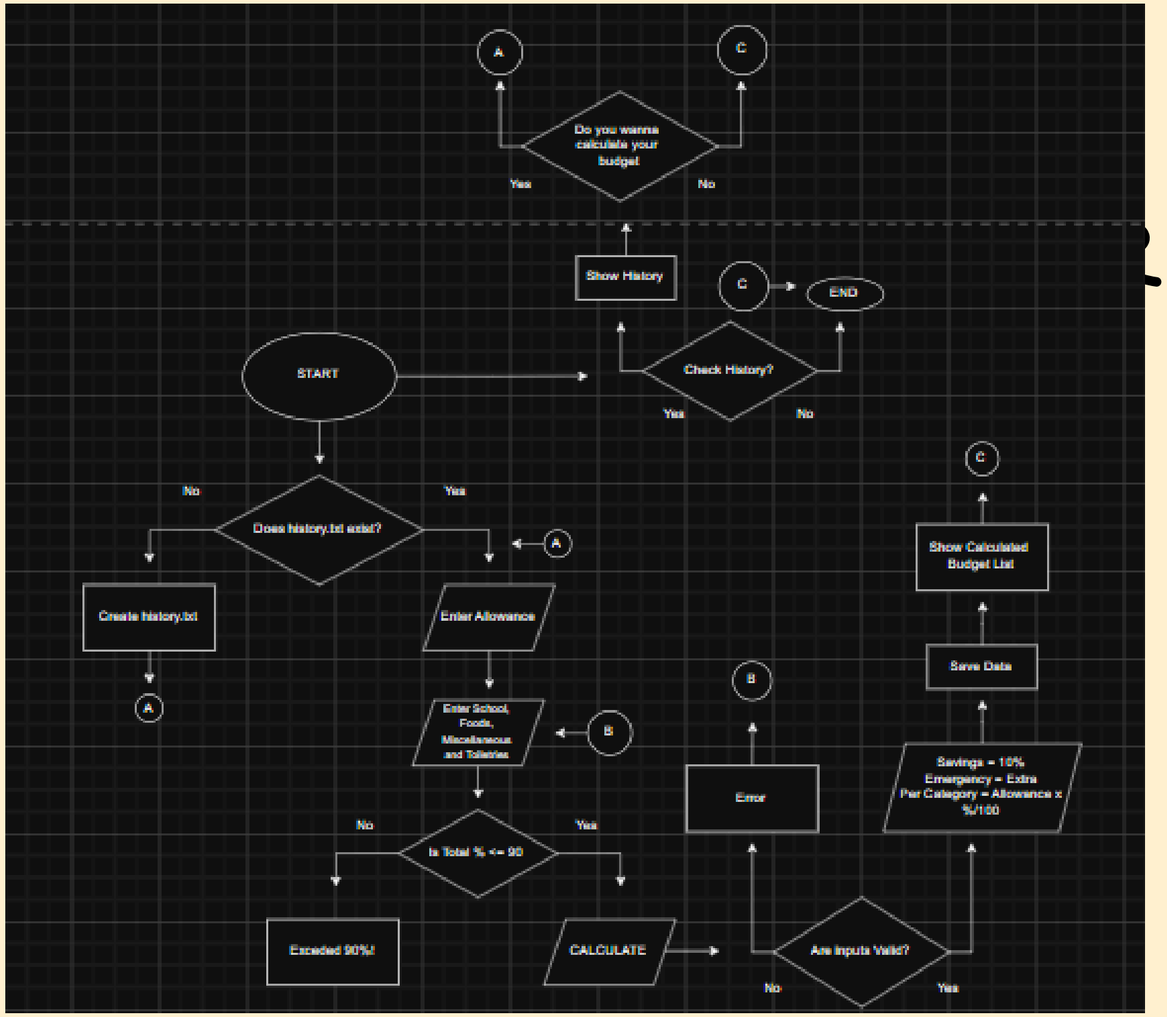
Methodology (simplified)

- GUI (Tkinter) – used to create the window, buttons, and input spots
- Input Validation – uses a validation command to make sure that the users can only type numbers, so that it would be easier.
- The 90/10 Rule – this rule ensures that 10% is automatically allocated to savings for unexpected situations.
- Event Binding – Uses .bind("<KeyRelease>") so that the live_update function triggers every time a student types a number.
- File I/O – makes use of Python's os and open() function to append data for long-term storage

Navigation Flow

1. First, the user launches SmartSpend.
2. Second, the user enters their total Allowance.
3. Third, the user enters percentages for different categories.
4. Fourth, the system checks if the total percentage exceeds 90%.
5. Next, the system converts percentages to numbers and adds leftovers to Emergency Funds.
6. Then, the results are displayed via an animation and saved to history.txt.
7. Finally, the user can click "Show History" to view past budgets.

Flow Chart


A
C
Do you wanna calculate your budget
Yes
No
Show History
C
END
START
Check History?
Yes
No
C
No
Yes
Does history.txt exist?
A
Show Calculated Budget List
Create history.txt
Enter Allowance
Save Data
A
B
Enter School, Foods, Miscellaneous and Toiletries
B
Savings = 10%
Emergency = Extra
Per Category = Allowance x %/100
Error
No
Yes
Is Total % <= 90
Exceded 90%!
CALCULATE
Are inputs Valid?
No
Yes

Reflection:

Coding SmartSpend taught us how to turn a problem into a simple, user-friendly tool. One of the biggest challenges we had was that we didn't know how to use Ktinker, so we had to sacrifice a few hours of our weekends learning and making sure that our code works and doesn't crash.

But we also learned a lot of things, like how to fix our code efficiently, how to utilize resources effectively and just how fun the feeling is when you see that the code you've been working on for weeks works. We also learned that in these kinds of projects, perseverance, teamwork, and patience really helps.

Presentation #101

Thank You So Much

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